LCEC-IO — EtherCAT slave I/O board

Root sheet, milestone M3: Power (2) and MCU (3). Sheets 4 to 8 (EtherCAT, digital inputs, MPG, relay outputs, analog outputs) are added in M4 and M5; the full root sheet follows in M6.

Power

File: 02_power.kicad_sch

MCU

MPG_A
MPG_B
ESC_IRQ
ESC_SYNC0
ESC_SYNC1
ESC_MISO
DI[1..8]
SEL[1..12]
RELAY_OE
ERR_LED
ESC_CS
ESC_SCK
ESC_MOSI
ESC_RST
DAC_CS
DAC_SCK
DAC_MOSI
AO_CLAMP
DO[1..8]
CLK_ESC

File: 03_mcu.kicad_sch

No-connect flags on the MCU sheet pins are placeholders: they are replaced by wires to sheets 4 to 8 in M4 and M5.

Root sheet: milestone M3 holds the Power and MCU sheets; sheets 4–8 follow in M4 and M5

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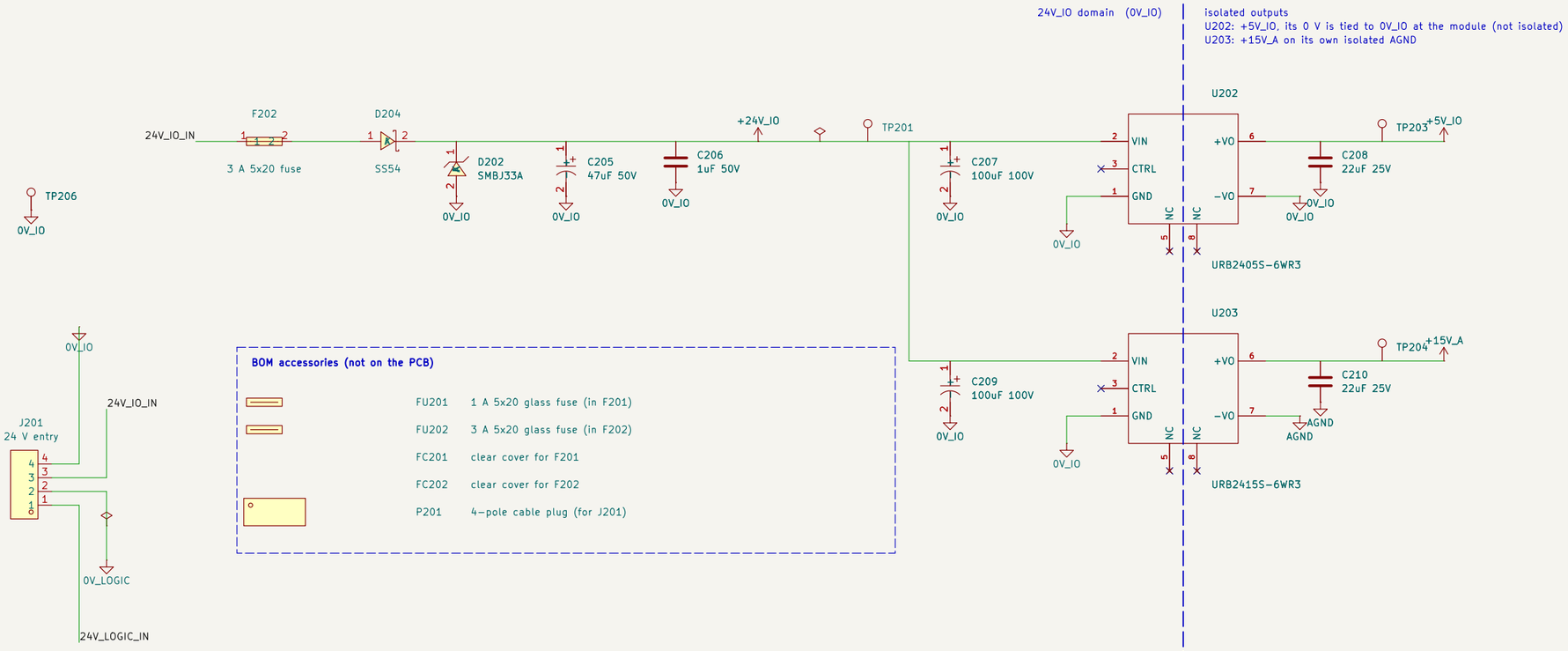
Sheet: /
File: lcec-io.kicad_sch

Title: LCEC-IO — EtherCAT slave I/O board

Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/3

2 POWER – 24 V entries, protection, three isolated DC–DC modules

Input 24 VDC +/–20 % = 19.2 … 28.8 V. Chain per input: fuse > SS54 reverse–polarity diode > SMBJ33A TVS > 47 uF + 1 uF > 100 uF/100 V at each module (URB datasheet Cin).



Loads: +3V3 about 0.4 A of the module's 1.35 A (30 %). 24V_LOGIC input about 0.3 A at full module load: 1 A fuse.

24V_IO: 5 V module 7.3 W + 15 V module 7 W at full load = 0.75 A, plus relay coils 0.18 A and sensors via PTC: 3 A fuse.

URB input 9–36 V (40 V absolute max, 50 V for 1 s surge). SMBJ33A clamps 53.3 V at 11.3 A: see the PR for the proposed SMBJ30A (48.4 V).

CTRL pin left open = module on. NC pins are not connected inside the module.

Two 24 V entries, protection, three isolated DC–DC modules

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Sheet: /Power/

File: 02_power.kicad_sch

Title: LCEC–IO – 2 Power

Size: A3	Date: 2026–11–06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/3

3 MCU — STM32F405VGT6, shared 25 MHz clock, SWD, UART2, heartbeat LED

Pin map: docs/plan.md (MCU sheet). 51 signals used, PH1 is a test pad, the 30 pins of ports D and E are free for future use (no-connect flags).

Firmware: HSE bypass with the 25 MHz oscillator, PLLM 25 / PLLN 336 / PLLP 2 = 168 MHz. IWDG enabled in option bytes (fail-safe).

X301
+3V3
GND_L
OE
Gnd
Vdd
Output
25 MHz
C313
100nF
R301
22R
CLK_MCU
R302
22R
CLK_ESC

OE high = oscillator running. 22 R series resistors: PH0 (HSE bypass) and LAN9252 OSCI (sheet 4).

100 nF at VBAT and at each VDD pin, drawn beside the pin it serves

C308 4.7uF
C301 C302 C303 C304 C305 C306 C307
FB301 600R@100MHz
VDDA
C309 1uF
C310 100nF

U301
STM32F405VGT6

NRST
BOOT0
CLK_MCU
PH1_TP
MPG_AD
MPG_BD
ESC_IRQD
ESC_SYNC0D
ESC_SYNC1D
DI[1..8]
SEL[1..12]

PH0/OSC_IN
PH1/OSC_OUT
PB4/TIM3_CH1
PB5/TIM3_CH2
PC4/EXTI4
PA0/EXTI0
PC13/EXTI13
PC0 PC1 PC2 PC3 PC6 PC7 PC8 PC9
PB0 PB1 PB2/BOOT1 PB8 PB9 PB10 PB11 PA15 PB6 PB7 PC14/OSC32_IN PC15/OSC32_OUT
PE0 PE1 PE2 PE3 PE4 PE5 PE6 PE7 PE8 PE9 PE10 PE11 PE12 PE13 PE14 PE15

PA8 PA9 PA10 PA11 PA12 PC10 PC11 PC12
DO1 DO2 DO3 DO4 DO5 DO6 DO7 DO8
DO[1..8]

PD2 PA1 PD3
HB_LED
RELAY_OE
ERR_LED

PA4/SPI1_NSS PA5/SPI1_SCK PA6/SPI1_MISO PA7/SPI1_MOSI PC5
ESC_CS ESC_SCK ESC_MISO ESC_MOSI ESC_RST

PB12/SPI2_NSS PB13/SPI2_SCK PB15/SPI2_MOSI PB14
DAC_CS DAC_SCK DAC_MOSI AO_CLAMP

PA13/SWDIO PA14/SWCLK PB3/SWO PA2/USART2_TX PA3/USART2_RX
SWDIO SWCLK SWO UART2_TX UART2_RX

SWD (Cortex 10-pin): 1 VTref, 2 SWDIO, 3/5/9 GND, 4 SWCLK, 6 SWO, 7/8 not used, 10 NRST

PD0 PD1 PD4 PD5 PD6 PD7 PD8 PD9 PD10 PD11 PD12 PD13 PD14 PD15

J301
SWD 2x5 1.27 mm

USART2: 1 TX (PA2), 2 RX (PA3), 3 GND

J302
UART2_TX
UART2_RX
USART2

Heartbeat LED (PA1)
R307 1k
LED301 heartbeat

VSS VSS VSS VSS VSSA VCAP_1 VCAP_2
C311 2.2uF
C312 2.2uF

VCAP_1 and VCAP_2: 2.2 uF each (low ESR)

Boot, reset and fail-safe pull resistors

BOOT0 R303 10k
NRST C314 100nF
RELAY_OE R304 10k
+3V3 R305 10k AO_CLAMP
+3V3 R306 10k ESC_RST

Fail-safe: PD2 low = relays off, PB14 high = analog outputs clamped, PC5 high = LAN9252 not in reset.

Test points: SPI1 (LAN9252), SPI2 (DAC), relay OE, analog clamp, MPG A/B and the spare pin PH1

TP301 ESC_SCK
TP302 ESC_MISO
TP303 ESC_MOSI
TP304 ESC_CS
TP305 DAC_SCK
TP306 DAC_MOSI
TP307 DAC_CS
TP308 RELAY_OE
TP309 AO_CLAMP
TP310 MPG_A
TP311 MPG_B
TP312 PH1_TP

STM32F405VGT6 (LQFP-100), shared clock, SWD, UART, heartbeat LED
Yantra Innovation — LCEC-IO
Sheet: /MCU/
File: 03_mcu.kicad_sch
Title: LCEC-IO — 3 MCU
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